Zihan Wang

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Education

Courant Institute of Mathematical Sciences, New York University

Ph.D. in Mathematics 09/2024-

- Advisor: Qi Lei.
- Honors: Moses A. Greenfield Research Prize.

Courant Institute of Mathematical Sciences, New York University

M.S. in Mathematics 09/2022-05/2024

- Honors: Glenn Y Louie Endowed Fellowship.

School of Mathematical Sciences, Peking University

B.S. in Statistics 09/2018-07/2022

Research Interests

Understanding large language models and deep learning, mainly from theoretical perspective, topics include chain of thought, self-improving models, optimization theory and machine learning privacy. Also interested in AI for math.

Selected Pubulications & Preprints (* represents alphabetical order)

1. Sheng Liu*, **Zihan Wang*,** Yuxiao Chen, Qi Lei. “Data reconstruction attack and defenses: A systematic evaluation”, AISTATS 2025.
2. **Zihan Wang**, Arthur Jacot. “Implicit bias of SGD in L2-regularized linear DNNs: One-way jumps from high to low rank”, ICLR 2024 spotlight.
3. **Zihan Wang**, Jason D. Lee, Qi Lei. “Reconstructing Training Data from Model Gradient, Provably”, AISTATS 2023.

Invited Talks

1. Jun. 2023. Reconstructing Training Data from Model Gradient, Provably.
   Machine learning reading group, University of California Berkeley, Berkeley.

Recent Projects

Self-improving generation models via verifier-guided closed-loop post-training

- Developing a **self-improving generation model** trained on model-generated data with verifier-guided selection and weighting.
- Built an experimental framework to compare off-policy vs on-policy self-improve updates.
- Apply the closed-loop method to constructions in mathematics.
- **Status:** in preparation.

Two-timescale optimization theory for mixture-of-experts

- Developing a two-timescale **training theory for softmax-router mixture-of-experts.**
- Established key martingalel arguments and dynamical system style analysis for inner loop and outer loop, which lead to convergence results.
- **Status:** in preparation.

Understanding Chain-of-Thought with Markovian perspective

- Modeled multi-step reasoning traces as a Markov process and identified **“transition alignment” as the main structural** condition under which chain-of-thought improves performance.
- Proved that when step-to-step behavior is stable and consistent, chain-of-thought can reduce inference-time sampling cost roughly proportional to the number of reasoning steps.
- Verified the theoretical results with controlled experiments, where unrelated factors are eliminated.
- **Status:** manuscript under review.

Systematic evaluation of reconstruction attacks and defenses

- Recast **data reconstruction as an inverse problem** to enable apples-to-apples comparison of attacks and defenses.
- Derived algorithmic upper bounds and matching information-theoretic lower bounds for reconstruction error in two-layer neural networks, clarifying how difficulty scales with dimension and model size.
- Proposed a utility-matched evaluation protocol and developed a stronger attack under this metric to reassess common defenses.
- Empirically compared different attacking methods and defending strategies with realistic dataset.
- **Status:** accepted by **AISTATS 2025.**

Implicit bias of SGD in matrix completion problem

- Proposed an attraction mechanism of the **implicit bias of SGD** towards low rank solutions: the dynamics can move from higher-rank minima to lower-rank minima, while the reverse transition is not possible.
- Analyzed L2-regularized deep linear networks for matrix completion, where different local minima correspond to solutions with different effective ranks.
- Verified the theoretical results with numerical experiments.
- **Status:** accepted by **ICLR 2024** as a spotlight.

Reconstructing training data with single model gradient in federated learning

- Proposed a tensor decomposition based method of data reconstruction from only one model gradient at random initialization.
- Mathematically derived the error bound of the method and improved its dependence with data dimension under 2-layer fully connected network setting.
- Empirically verified the method on toy examples and MNIST dataset.
- **Status:** accepted by **AISTATS 2023.**

Service

- Reviewer in conferences including ICLR, ICML, NeurIPS and AISTATS.

Availability

May 21st to August 31st.